

Towards Unconstrained Speech Decoding from ECoG

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Background

- **Goal:** Decode speech from ECoG by converting neural activity into phoneme sequences. The long-term aim is to support long recordings, identify when speech occurs, and decode only those segments.
- **Prior Work.** Existing approaches include linear decoders, LSTMs, CNNs, Transformers, and end-to-end speech neuroprosthesis systems that map ECoG to phonemes, text, or audio (Table 1).

Model Type	Lab / Year	Task	Accuracy	Data Availability
Linear Decoder	Cogan Lab (Duke), 2023	Phoneme classification (9 categories)	~45% (chance = 11%)	Very limited, high-resolution
LSTM	Cogan Lab (Duke), 2023	Decode 3-phoneme sequences	46–51%	Very limited (4 participants)
3D CNN	Flinker Lab (NYU), 2024	Brain-to-speech reconstruction	0.80 correlation	Large (48 participants, ~400 trials each)
3D Transformer	Flinker Lab (NYU), 2024	Brain-to-speech reconstruction	0.80 correlation	Large (same dataset as above)
RNN + Language Model	Willett et al., 2023	Brain-to-text	86% word accuracy, 62 WPM	Massive (10,850 sentences) + strong LM
RNN (streaming)	Chang Lab, 2025	Real-time silent-speech decoding	68–90% word accuracy	Large + strong LM
LSTM + Audio Synthesis	Chang Lab, 2024	Brain-to-speech (from 20 min data)	72% human word recognition	Minimal (20 min) + pretrained audio models
Transformer (real-time)	Stavisky Lab, 2024	Real-time brain-to-speech	0.83 correlation; 56% word accuracy	Massive (months of chronic recordings)

- **Limitations.** The highest reported accuracies (86% word accuracy in Willett 2023; 68–90% in Chang 2025) rely on privileged conditions: thousands of repeated sentences from one participant, stable electrodes over months, limited vocabularies, and strong language-model correction. In less constrained settings—limited data, variable participants, irregular speech timing—accuracy drops substantially (30–50% phoneme accuracy, 10–58% word error rate). High performance in unconstrained, real-world scenarios remains unsolved.
- **Data Sources.** I will use two complementary datasets:
 - Cogan Lab (Duke): High-resolution micro-ECoG with fine spatial granularity (1.3–1.7 mm electrode spacing), enabling precise localization of neural activity.
 - Stavisky Lab (Stanford): 15 months of continuous recordings from a single participant, enabling models that must operate over extended time spans.

Objectives

- **Fine-tune** foundational models across architectures, with a focus on Mamba, which has not been applied to neural speech decoding
- **Develop** a hybrid CNN-Mamba architecture for long-range sequence modeling
- **Adjust** model architecture to ensure robust decoding across extended recordings with intermittent speech

Data

- Source: Cogan Lab (Duke)
- Collection: Micro-ECoG recordings during production of nonsense syllables
- Quantity: 8 patients, 144 trials/patient, 3 phoneme/trial, 9 possible phonemes
- Format: 200 time points (± 500 ms around phoneme onset) per sample

Model

PCA + SVM

- Classifies one phoneme at a time from a single neural recording segment
- Structure:
 - PCA: Reduces high-dimensional ECoG to components capturing most variance
 - SVM (RBF kernel): Draws nonlinear decision boundaries between phoneme classes
- What I did: Built and trained entire pipeline from scratch

Seq2Seq RNN (Cogan et al., 2023)

- Classifies three consecutive phonemes from one continuous recording
- Structure:
 - Temporal CNN: Extracts local time patterns from raw ECoG
 - Encoder GRU: Compresses entire time window into one summary vector
 - Decoder GRU: Generates three phoneme predictions sequentially
- What I did: Used existing architecture, tuned hyperparameters

MIRepNet (Wu et al., 2025)

- Classifies one phoneme at a time
- Structure:
 - Temporal Conv + Spatial Conv: Extracts time and channel patterns
 - Transformer Encoder (6 layers): Learns long-range dependencies via self-attention
 - Classification Head: Maps to phoneme classes
- What I did: Froze all conv + transformer layers; finetuned classification head

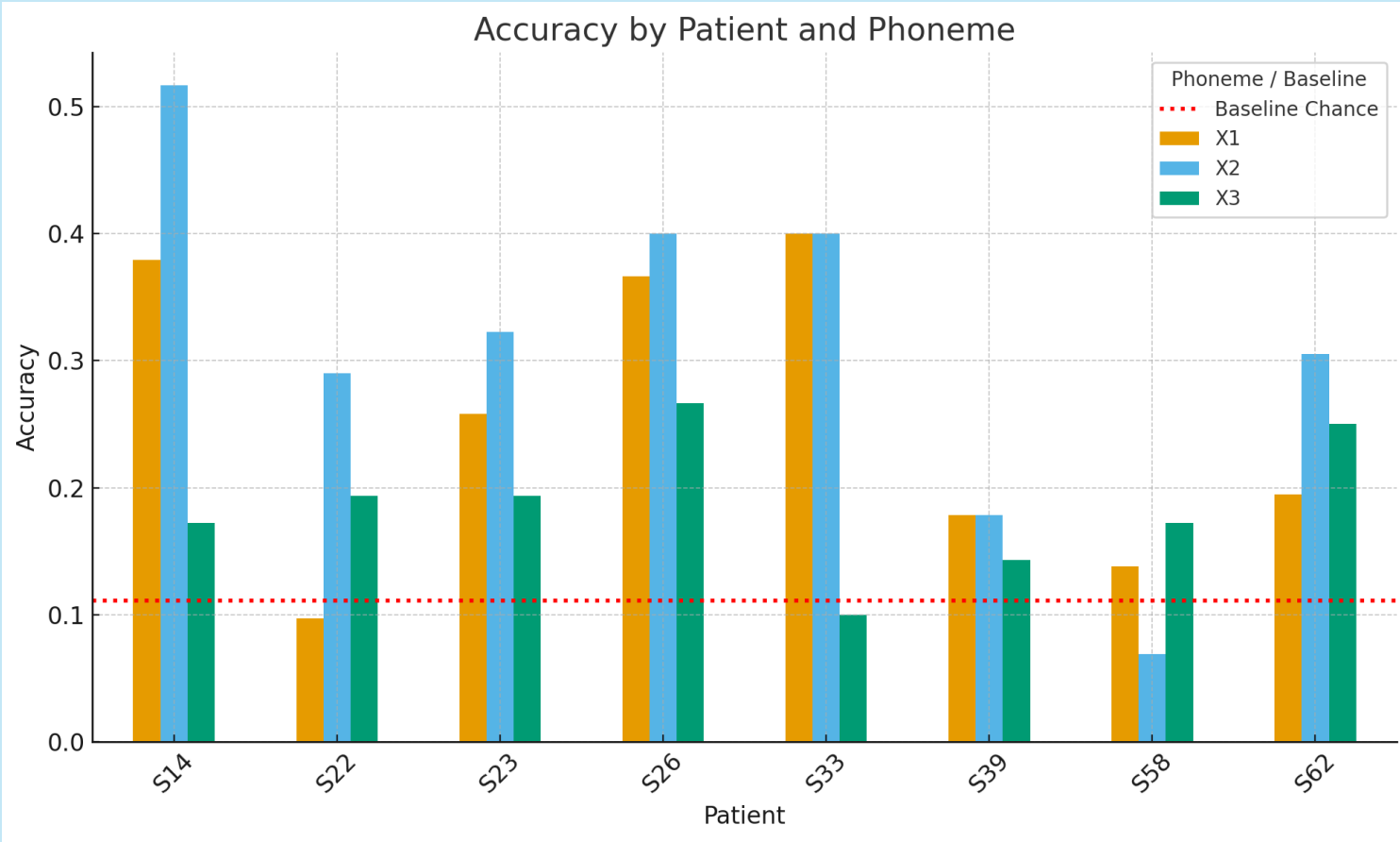
Spatiotemporal CNN (Flinker et al., 2024)

- Classifies one phoneme at a time
- Structure:
 - 3D CNN + Residual Blocks: Extracts spatiotemporal features across electrode grid and time
 - Projection: Produces compact latent representation
- What I did: Froze entire encoder; finetuned new classification head only

Result

PCA + SVM (Linear Baseline)

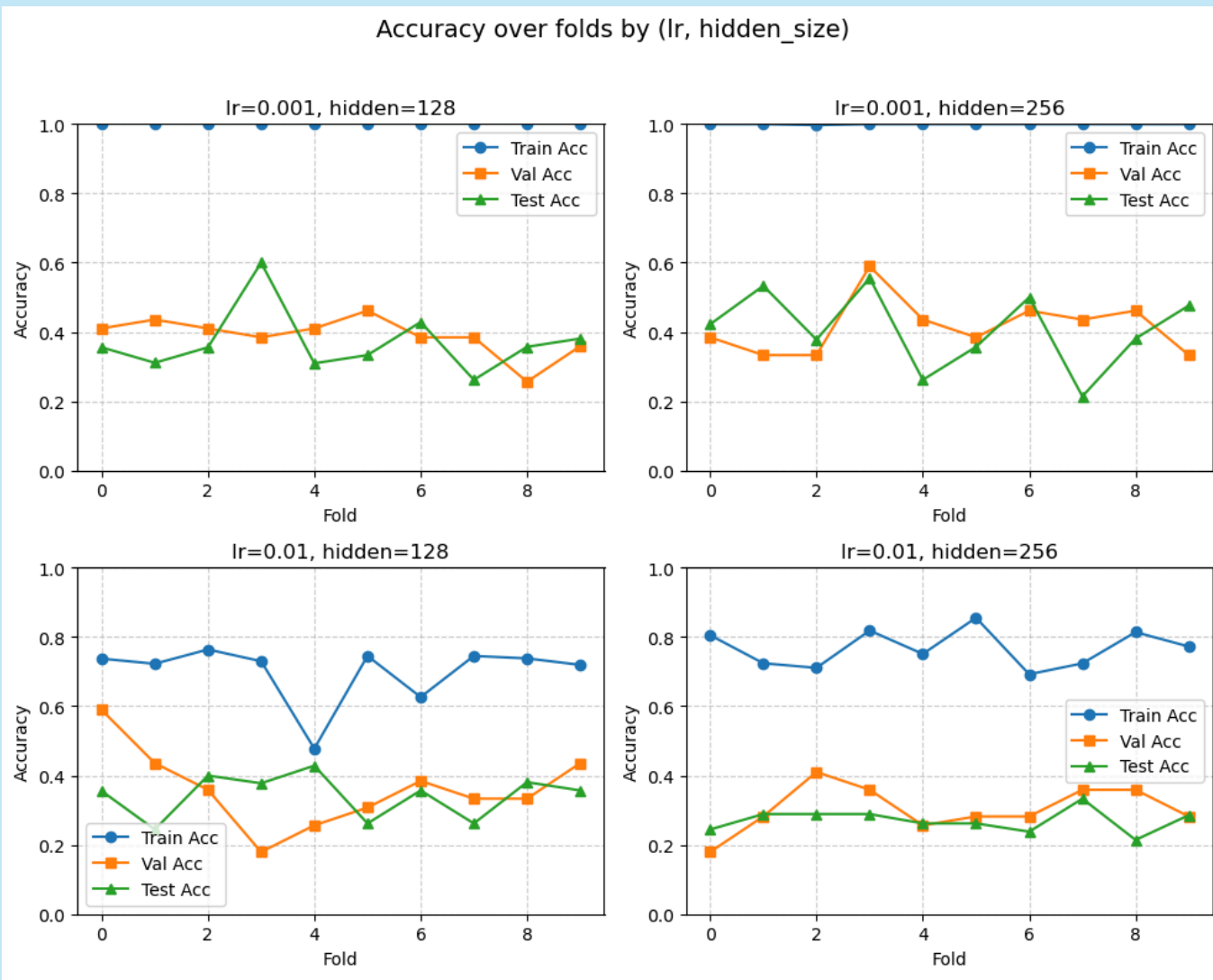
- Accuracy across 8 patients $\times$ 3 phoneme positions:



- Range: 7–52%, varying by patient and phoneme position
- Position 2 consistently highest (middle phoneme better aligned to window)

Seq2Seq RNN (Cogan et al., 2023)

- Hyperparameter sweep: 2 learning rates $\times$ 2 hidden sizes $\times$ 10 folds = 40 runs



- Training accuracy: 70–100% (memorization)
- Validation accuracy: 17–46% (unstable, limited data)
- Test accuracy: 21–60%

MIRepNet (Wu et al., 2025)

Scenario	Accuracy	Interpretation
Train on all 8 patients, test on held-out trials	~12%	Cross-patient generalization fails (chance = 11%)
Train on one patient, test on same training data	100%	Model memorizes small dataset
Train on one patient, test on unseen trials from same patient	55%	Some generalization, but overfitting limits performance

Future Directions

- Finetune a MI classification MAMBA model for single phoneme neural speech decoding
- Switch to the Stavisky dataset with 15 months of recordings from one patient
- Build a hybrid CNN + Mamba architecture and train it
- Extend to continuous speech decoding on minutes-long recordings with automatic speech detection